



ADAPTATION FUND



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Project

Addressing climate change risks to farming
systems in Turkmenistan

BRIEF DESCRIPTION

The proposed project aims to overcome the above barriers to addressing immediate and long term adaptation needs in the water sector in Turkmenistan in order to achieve greater water efficiency and productivity under climate change induced aridification. The project will therefore aim to strengthen water management practices at national and local levels in response to climate change induced water scarcity risks to local farming systems in Turkmenistan. The project takes a comprehensive approach towards achieving this objective by encompassing national level water policy and local community level action to improve water efficiency and supply services.

United Nations Development Programme in Turkmenistan

Project Title:

«Addressing climate change risks to farming systems in Turkmenistan at national and community level»

Project Duration: 5 years

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Ministry of Nature Protection of Turkmenistan

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United Nations Development Programme (UNDP)

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«Addressing climate change risks to farming systems in Turkmenistan at national and community level»

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SITUATION ANALYSIS

Introduction

Climate change is projected to have significant impacts on water resources in an already arid Turkmenistan. Water availability and supply are likely to suffer from increasing shortages due to elevated temperatures, overall climate aridification and competition for water arising from regional trans-boundary water issues. Turkmenistan's inherent aridity and reliance on agriculture as a source of both income and food renders the country particularly vulnerable to these climate change impacts.

In response, the Government of Turkmenistan is proposing to take a comprehensive approach to water adaptation in the agricultural sector. This initiative, which seeks financing through the Adaptation Fund, aims to increase resilience in 3 different agro-climatic zones in Turkmenistan by implementing hard water efficiency and irrigation measures, and to develop water user associations benefiting over 30,000 farmers. These communities are among the most vulnerable and water stressed, and for the most part lack access to state subsidy or support. To ensure that the beneficial impacts of the project can be replicated among other communities, the project will also seek to internalize climate change risks into water policies at the national level by establishing price incentives, technologies and management systems to achieve greater water use efficiency. It focuses on increasing the resilience of water resources for the most vulnerable and water-stressed communities, who are engaged in non-state agriculture, horticulture and livestock management and who are unlikely to benefit from Government's large scale water supply and storage infrastructure. The programme contributes directly to the AF's portfolio level objective *«to increase adaptive capacity to respond to the impacts of climate change, including variability at local and national level»*.

Current State of the Agricultural Sector

The agriculture sector of Turkmenistan is hugely capital and labor intensive. The agricultural industry is mainly owned and controlled by the state, with a few private producers and farm businesses starting to emerge in livestock, agricultural and processing sectors over the last decade. The main crops mandated by the state are cotton and wheat, as well as smaller amounts of rice and sugar beet. Almost all public investment is directed to production of these two strategic crops, based on an economic policy of self-sufficiency in grains and maintaining the export potential for cotton products. This policy has greatly affected the structure of the agricultural sector and its potential for production, since thousands of hectares of land under orchards, horticulture and fodder crops have been diverted to production of winter wheat.

Over recent years, the Government has begun to recognize the need to diversify the agricultural sector. This process has been driven by increasing frequency of drought in agricultural producing regions, and the associated economic losses experienced by the state. For the growing season of 2010 the state has allocated some land for crops other than state or government mandated crops to be grown in each of the five provinces based on soil-climatic conditions in these regions. As a result of this, each province has been allocated land on a leasehold basis for growing maize, barley, lucerne and other forage crops, a practice aimed at promoting crop rotation and thus improving soil quality. The government is also beginning to invest in high efficiency irrigation technology for water intensive crops, to include (subsoil) drip irrigation systems, mobile sprinklers and waste water drainage and recycling.



In accordance with NASA satellites' data the Sun has become still nearer to the Earth.
The cloud layer height has gone down at 1 percent, i.e. at 30-40 meters, for the last decade.

The **livestock** subsector is dominated by the private sector, with more than 80% of all products produced by private farmers following the decommissioning of Soviet state livestock farms. Private rural households operate on a lease agreement, according to which the leaseholder provides feed and veterinary services and makes sure that the livestock is kept in good condition. Based on the lease agreement the leaseholder is allocated land for growing fodder crops and receives all the livestock products produced and half of the offspring during the lease period. This lease arrangement predominantly applies to the lease of cattle, whereas sheep and goats are mainly kept by the state association itself due to relatively easier husbandry and production conditions. Despite a lack of state investment, livestock management has managed to develop a certain degree of efficiency and productivity, and this subsector now provides a good model for further private sector development within the agricultural sector.

The **fruit and vegetable** subsector is the most independent of the agricultural sector in Turkmenistan with almost 100% of production generated privately both by independent farmers and leaseholders. The total land area that is used for production of small scale farming is negligible compared to the grain, cotton or livestock subsectors, but nevertheless it is highly productive and the most economically viable within the country's agricultural sector. Climatic conditions have historically allowed the country to produce high quality fruit and vegetable products and prior to collapse of the Soviet Union, Turkmenistan was a major exporter of fruit and vegetables to northern parts of the Soviet Union. Over recent years, land allocated to produce winter wheat was increased almost five times largely at the expense of areas dedicated to feed, fruit and vegetable crops. Prices and availability fluctuate between seasons, reflecting an undeveloped processing sector and lack of appropriate technology, facilities and infrastructure for cool storage. Despite the emergence of some private sector activity, this remains a key area for development, and demand for locally sourced product in the markets remains strong.

Climate change impacts:

Meteorological drought is a semi-permanent condition in Turkmenistan. The country receives on average only 191 mm of precipitation per year. The country is therefore inherently water scarce, characterized by a continental and very dry climate, with low levels of precipitation and moisture (35% on average). Turkmenistan is a predominantly arid country with over 80% of its territory characterized by desert and oases. Water shortages and periods of drought are common, a situation which is likely to be exacerbated by climate change with consequences for development, economic growth and livelihoods. Almost half of the population is employed in the agriculture sector, and approximately 55% reside in rural areas.

Climate observations show that the air temperature is steadily increasing in Turkmenistan as in the whole of Central Asia. Precipitation will become more variable, with increased frequency and intensity of drought and flood spells. Glacial retreat in Pami-Altai will have significant impacts on water flows of the Amu Darya River¹. As a result, significant decreases in water supply and agricultural production are expected. It is estimated that 30% of glaciers feeding the waters of Turkmenistan have already been lost during the past century, as a result of global warming. This is particularly alarming for the country whose water runoff formation is fully dependent on natural flow from glaciers. A trans-boundary river, the Amu Darya is the main source of irrigation for a number of countries in the region including Uzbekistan. The planned development of upstream hydro-electric projects in Tajikistan threatens the potential stability of downstream flows. The expected 15% reduction in flow of the Amu Darya by 2030 will have dramatic impacts on agriculture and food production in Turkmenistan. Other river flow rates are expected to decline at even faster rates (up to 30% reduction). At present, agriculture consumes 92% of all surface waters available in the country (2% – communal and 6% – industry). The situation becomes even more critical with the anticipated impacts of climate change, which will be characterized by increasing temperatures and evapotranspiration rates and decreasing precipitation

¹ The First National Communication to UNFCCC, Turkmenistan, 1998



Levels. As a result, the average reduction in run off rates in terms of surface water collected in national storage and distribution systems is expected to be 10%, whereas during vegetation periods the reduction in run off rates will reach 30 - 40%.

In summary, predicted climate change impacts include:

- *An increase in average annual temperature of between 4.2 and 6.1°C by 2050², which will include an increase in the number of extremely hot days (i.e. days over 40°C)³;*
- *A reduction in annual average rainfall of between 15 and 56% by 2050;*
- *An increase in average regional evaporation rates of 48% by 2050ⁱ;*
- *An increase in the frequency and intensity of drought and flood⁴ spellsⁱⁱ;*
- *A 15% reduction in flow rates for the Amu Darya;*
- *A 30% reduction in flow rates for other river systems.*

In the context of the above additional pressures, the following are critical underlying causes of vulnerability:

(i) Deteriorating irrigation infrastructure and subsidized water prices.

The availability of water in Turkmenistan is already constraining development and will do so even further in the face of climate change. In fact, water consumption per capita in Turkmenistan is more than twice that of any other country in Central Asiaⁱⁱⁱ. However, the high water consumption levels are largely related to the inefficiency of irrigation systems in the country, as opposed to high household consumption. Indeed, some 28% of Turkmen are without access to potable water sources^{iv}. Tariffs are set by the government on a below cost recovery basis. This is done primarily to mitigate the social impacts of market pricing, and to ensure that those populations with a relatively modest income base are not disadvantaged. Such an approach makes the current water systems financially unsustainable and as such, this dampens the interest of the private sector to invest in the absence of compensatory financial mechanisms and economic instruments. As a result, incentives for water efficiency are largely absent, thus large farmers use water inefficiently, and the quality of local service delivery for small holders suffers.

(ii) Allocation of water resources to irrigate intensive cash crops due to historical reasons related to the Soviet period.

The agricultural sector is the main consumer of water within Turkmenistan. Agriculture is a critical sector of the economy accounting for almost one-fifth of GDP and is a source of livelihood for half of the population. Turkmenistan took an initial step in 1997 – 1998 in changing the status of most farmers to “lease-holders.” However, in practice, the rural economy continues to operate primarily under state control, with the government controlling both inputs and providing a market for produce for strategic crops (cotton, wheat, rice, and sugar beet). Virtually all cotton and wheat crops are grown under the system of state mandate and procured by the state at below-market prices. Some initial positive steps to initiate reforms of this system for cotton have been recently taken by the government. To improve the productivity of these crops, the government provides some incentives to farmers. These incentives are mainly provided to commercial farmers (who are involved in the large-scale production of wheat, cotton or rice) and not to the rural poor (who rely largely on subsistence farming of grains, melons and vegetables, or local markets). Unlike commercial farmers, the rural poor are unable to afford pumps for water and hence their productivity suffers. Thus, the current water policies burden the state budget and

² These estimates are based on the findings of five general atmosphere and ocean circulation models (GCM) reported in Turkmenistan's Initial Communication on Climate Change (1998). The GCM with the most plausible results on temperature predictions was the UK89 model (equilibrium model of the United Kingdom Meteorological Agency). According to this scenario, temperature is predicted to increase by 5.5°C by 2050.

³ The GDFL model scenario (equilibrium model of Geophysical Fluid Dynamics Laboratory, University of Princeton, USA), however, predicted no change in rainfall (Turkmenistan's Initial National Communication on Climate Change, 1998).

⁴ Floods are uncommon in Turkmenistan but they do still pose a threat to communities and infrastructure (see: <http://www.preventionweb.net/english/countries/statistics/risk.php?cid=178>).

ⁱ Turkmenistan: Initial National Communication on Climate Change, 1998.

ⁱⁱ Turkmenistan Country Analysis. United Nations, 2008.

ⁱⁱⁱ Central Asia Regional Risk Assessment: Responding to Water, Energy and Food Insecurities. UNDP Regional Bureau for Europe and CIS. New York. 2009.

^{iv} Central Asia Regional Risk Assessment: Responding to Water, Energy and Food Insecurities. UNDP Regional Bureau for Europe and CIS. New York. 2009.



The biggest hailstone witnessed by people fell at Coffeyville (Kansas, USA).
It weighed nearly 700 grams.



do not free up resources for service improvement to farmers, especially local small holders. There is a significant opportunity cost. At the same time, farmers involved in large scale production of water intensive crop varieties do not receive adequate price signals to use water more efficiently. Given the increasing water shortages and priorities assigned to cash crop production the small holder subsistence farmers bear a disproportionate burden of intensifying water deficits.

The government is increasingly aware of climate change related pressures on water availability and agricultural productivity. The Second National Communication is currently awaiting approval at the Cabinet of Ministers. For example, planned cotton production targets in Turkmenistan only reached 79% in 2 000 and 63% in 2001, largely due to the severe and unprecedented hydrological drought. Water requirements for crops will rise 30 – 40% due to expected higher evaporation rates. Unless the efficiency of irrigation systems climbs from 57% at present to 75% by 2050, there will be a water deficit of 14km³ in irrigated agriculture. Humidity deficits will also impact productivity of pastures⁵. This is an alarming indicator that the currently heavily subsidized system of water supply is almost certainly unlikely to be sustainable against rising water deficits resulting from climate change.

The government is also concerned that rising water deficits disproportionately impact the poorest farmers, who are among the most vulnerable communities. In this regard, the importance of rural development and social sectors has been underscored by the current government, and it has recently pledged significant resources⁶ toward these priorities. A recent assessment of Investment and Financial Flows (I&FF) for government adaptation activities in the water sector identified critical measures such as the introduction of economic instruments and financial mechanisms to make more water sector services self-sustaining under a changing climatic baseline. However, such instruments/mechanisms need to be carefully designed and applied. The assessment also identified the need for fundamental capacity development in key institutions in this critical sector. In addition, further recommendations suggested reaching out to the most vulnerable communities in various agro-ecological zones to tailor and implement local adaptation measures that improve access to water and promote livelihood resilience. The Adaptation Fund (AF) financed project proposed here seeks to implement such priorities as identified and prioritized by government. The proposed project aims to overcome the above barriers in order to achieve greater water efficiency and productivity under climate change induced aridification. The project will therefore aim to strengthen water management practices at national and local levels in response to climate change induced water scarcity risks to local farming systems in Turkmenistan. The project takes a comprehensive approach towards achieving this objective by encompassing national level water policy and local community level action to improve water efficiency and supply services.

The government of Turkmenistan has come to realise that water is one of the key driving forces for its economic development, and under conditions of increasing scarcity water infrastructure needs to be upgraded to minimise losses to the system. The government therefore has dedicated significant budget allocations for technological upgrades in water infrastructure. Currently, this is primarily related to improving supply side delivery. This focuses on the upgrade of pumping stations and lining of canal systems. The largest investment has been in the construction of an artificial lake, Altyn Asyr, and associated canal infrastructure to collect drainage water, which will eventually be used for irrigation purposes following natural purification. This water is likely to benefit state farms who participate in the irrigation programme for government order crops (cotton, wheat etc.) in the Karakum region. The Government understands, however, that an approach based on supply side infrastructure may not be sufficient. It has begun a large scale investment in high efficiency irrigation technologies, including (subsoil) drip irrigations systems, mobile sprinklers, and waste water capture and reuse, and has committed to scale up these activities in the new Agriculture and Water Strategy. Because prospects for development of new supplies are limited in Turkmenistan, improvements in efficiency are paramount for reliable supplies during the periods of hydrological drought in more arid conditions to be brought

⁵ World Bank, Drought Management and Mitigation Assessment for Central Asia and the Caucasus, Regional and Country Profiles and Strategies, 2006

⁶ More than US\$4 billion has been pledged. However, it still remains to be seen how much will actually materialize.

Where do plants take water from in desert, indeed? Sands turn precipitation moisture into water at night. Deep underground waters, which long-root plants reach to, are considered to be another water source. But moisture is too little anyway, and desert plants have been developing leaf little evaporation capability for lots of millenniums.



about by climate change. Currently, the government plan does not address the water needs in the non-state agriculture sector. For this reason, the AF programme will focus on the development of community level water management approaches in those regions where there is significant potential for diversified non-state agriculture, horticulture and livestock management. It will also seek to ensure water availability for the non-state sector by addressing the lack of fiscal incentives for more efficient water use in the state sector, and developing progressive tariffs.

The project is structured so that the majority of its activities are at a community level to deliver concrete adaptation benefits to identified communities in three typical agro-pastoral regions (mountainous, desert and oasis). The project will work directly with selected communities to help improve their resilience to increasing aridity and water stress through identifying and implementing effective and locally acceptable adaptation measures. For example, improvements of local water management options and communal water delivery services, seasonal water rationing and more flexible payout mechanisms will be introduced. The project also reaches out to communities that do not benefit from the irrigation services and practice rain-fed agriculture. For them improvement of water retention and harvesting techniques are critical for long term availability of water. The project therefore aims to deliver local level improvements in water access and management. Adaptive capacity and improved adaptation policies at the national level will be complemented by the local adaptation actions in the three target regions. The project will be implemented in regions that represent typical conditions of three major agro-ecological zones in Turkmenistan – that is, mountain, desert, and oasis systems. The project will tailor locally appropriate adaptation measures in these three areas in order to improve water delivery services that are adequate and effective under the area specific circumstances (see the component 2 for further details).

In order to support the scale up of community level adaptation solutions, at a national level, the project will support the implementation of a series of legislative modifications, particularly to the water code, its subsidiary legislation and regulations. These changes will be informed by socio-economic impact assessments of climate change with cost-benefit analysis of adaptation measures. The assessments, and related capacity development efforts of local technical officers will allow policy makers to adjust water pricing schemes, set productivity targets and target those communities who are currently most disadvantaged. The project will help to apply progressive (graduated pricing) and differentiated water tariff that allows cross-subsidies across water users of varied categories (e.g. farmers engaged in commercial farming versus small holder farmers on marginal lands). Introduction of this policy will have strong implications on future improvements in water service delivery to more disadvantaged farmers who will benefit from improved services and cross-subsidization. In order to induce efficiency, the price signal must be fair. This means that the tariff applied must reflect actual consumption, measured in a reliable way. Water pricing is an important tool for resource allocation because it provides guidance, both to users and to planners, in comparing alternative solutions.



Legislative framework

The Water Code defines in more detail the functions of the Cabinet of Ministers in relation to water resource management and conservation. Particularly, the Cabinet of Ministers annually sets water use limits for each of the political-administrative units-five *welayats* and each *etrap* – as well as for each sector of the economy, and they define water extraction limits for each water source. Article 12 of the Law «On the Cabinet of Ministers of Turkmenistan» prescribes the Cabinet of Ministers to address at its meetings the measures on protection and conservation of land, water resources, plants and animal communities. The Water Code defines mechanisms for this in accordance with the existing legislative procedure regarding *Basin Schemes on Integrated Use and Protection of Water Resources*, as well as through national, multilateral, and regional programmes and projects. A recent analysis of water and land use legislation by UNDP indicated that while not promoting radical change, the opportunity exists for more detailed elaboration of implementation strategies to achieve the stated aim of improved water efficiency and associated increases in agricultural outputs. Progress in developing supplementary regulations, standards, and norms has been relatively slow but there is increased understanding within Government of the need to accelerate this process.



Baselines for UNDP intervention

UNDP has undertaken a number of initiatives in the water sector, including assessment of Investment and Financial Flows in relation to water sector adaptation that has informed the Second National Communication, and a study on institutional and policy mechanisms within the water sector. There are a number of parallel activities related to protected area management and biodiversity. Some of the identified sites for this project under component 2 are currently receiving support from a UNDP-GEF and GTZ co-supported medium-size project under the Ministry of Nature Protection on sustainable land management that is completing in 2010 (Capacity building and on-the-ground investments for sustainable land management; SLM-Project). This project is also reviewing the options for community participation approaches that deal with environmental issues (potentially to include climate change impacts). The Adaptation Fund (AF) project is seen as being complementary to the land management activities under the current project, addressing water stress on local community farming and pastoral systems. Some of the measures tested by the UNDP-GEF project (such as dune fixation and stabilization, restoration of vegetative cover) remain relevant for the AF project as well and lessons on best practices from that initiative will inform this one. However, the design of these measures, their spatial distribution and scale will be informed by the climate change risks that were not considered in the UNDP-GEF initiative. The AF project will address adaptive water issues, such as the introduction of watering points for pastoralists, innovative communal management modalities for water service delivery, such as seasonal water rationing and flexible payment options. It is expected that the established structures and UNDP-GEF track record in these pilot regions will result in lower implementation risks, and a high degree of engagement and goodwill from local partners. The project will coordinate closely with the UNDP Climate Risk Management programme for Central Asia which is currently in inception phase. All activities will be complementary.



STRATEGY

National priorities

The programme is fully in line with existing national priorities relating to water and agriculture. National policy is defined by the Water Code.

Its key provisions are;

- *Ownership over water and water resources of the country;*
- *Procedures of placing, design, and construction of water infrastructure;*
- *Types of water use and categories of water users;*
- *Procedures for receiving permits for specialized water use, rights and obligations of water users;*
- *Water protection arrangements and prevention of pollution including Water Protection Zones;*
- *State monitoring of water resources, State Water Cadastre;*
- *Participation of NGOs and wider public/communities organizations in water use and protection.*

The National Programme «The Strategy of Economic, Political, and Cultural Development of Turkmenistan Until 2020» sets out targets in relation to agricultural outputs. The Programme envisages an increase in agricultural production of more than 15 times only due to utilisation of the current natural resource and accelerated industrial potential. Wheat production is planned to grow 2.9 times and cotton production – 4.9 times by 2020 as compared with 2 000. A considerable proportion of irrigated agricultural lands is planned to be transferred to private sector enterprises. The private sector tenants will include joint-stock companies, daihan (farmer) cooperatives and unions. These categories of land users are expected to introduce more effective and efficient water use technologies and water saving practices.

A number of activities have recently cemented water and agricultural resilience as key strategic issues for national and regional development:

The water sector has been identified as the most important adaptation priority in the Second National Communication. This document will set out the projected water deficit under current climatic projections, and outline a number of policies and measures for the agriculture sector to mitigate impacts.

The Investment and Financial Flows (I&FF) assessment (2010) undertaken by UNDP and the Government of Turkmenistan identified the water sector as the primary area for adaptation investment. It concluded that in addition to ongoing programmes of supply side infrastructure investment and upgrade, it was important to develop an effective demand side response and establish management systems that would result in sustainable use and allocation of increasingly scarce water resources. The existing GoT five year water programme specifically promotes the more rational use of water. The National Socio-Economic Development Strategy of Turkmenistan to 2030 sets the objective of modernization of the water management system and improving water efficiency as well as increasing water storage capacity. The government has stated its aim of moving irrigated lands from a position of 90% state ownership to majority ownership by the private sector by 2020.



The National Agricultural Strategy specifically identifies the water requirements to meet state targets for strategic crop production. Recent legislation for the Agriculture sector - On Dayham Farm (2007) – sets out water provision requirements as central to the development of private sector agriculture, horticulture and livestock. There is an expected increase in the role of private farms, cooperatives and joint stock companies. The programme will support current national agriculture sector priorities, including growth in agricultural outputs, an increase in agricultural and pastoral productivity, development of agricultural processing, and more rational matching of land use potential with economic activity.

- *At recent meetings of the «Council of Elders», an ancient representing the traditional system of community leaders, in two consecutive years of 2009 and 2010, it was decided to make efforts to increase water productivity and reverse a trend of cultivated land expansion. This decision was triggered by increasing water deficits evidenced in the country in recent years due to climate change.*
- *In 2010, the Government organized an international forum on water technologies, including for the agricultural sector with a focus on demand side management and efficiency.*
- *The proposed concept is fully aligned with government's current priority on social protection and development. The community level interventions and communal management solutions for improved water efficiency and delivery services are in direct alignment with government priorities.*
- *Law of Turkmenistan “On State Support for Small and Medium Size Enterprise” (2009). This law supports the extension of private sector provision in the rural agriculture sector, including economic diversification into agro-processing industries.*

An integrated water and agriculture strategy is under development, and publication is expected in early 2011 upon approval by the Cabinet of Ministers. This strategy is being informed by a) The Programme of Agricultural Development to 2030, b) The Concept Note for Development of Water Economy to 2030, and c) Proposal for Development of Water Economy to 2030. The Strategy is being led by the Ministry of Agriculture with support from the Ministry of Water. The draft strategy is expected to recognize the key role of water in delivering growth in agricultural production, productivity, and yields, and promote both supply and demand side solutions to ensure the efficient use of water within the Agro-Industrial complex. The development of an integrated strategy is to be accompanied by the relocation of the agricultural, water and natural resource ministries, together with their supporting institutes into a single building.

In particular, the integrated strategy will set out a comprehensive plan to address water and agricultural productivity.

Some of the linkages under consideration include:

- *Integrated water resource management within agro-industrial areas*
- *Optimization modeling for agricultural production based on water use*
- *Improvements to existing irrigation systems through education and awareness*
- *Widespread introduction of drip and sprinkler irrigation technologies*
- *Development of additional reservoirs for agricultural irrigation purposes*
- *Development and introduction of drought resistant crops*
- *Reuse of drainage and waste water for agricultural purposes*

The Velna and the Nelba rivers close to Vongrovetz/Poland cross strictly right angularly. But, being not mixed at all, waters thereof follow their own courses further. The unique phenomenon is explained by differences in those rivers' water temperatures, speeds and stream levels.



Capacity development

The project is structured to allow a high proportion of funds to flow into capacity building activities and demonstrations of low cost resilient technologies and management structures at the macro level, particularly in component 1 and 3. As such, the components are expected to result in a higher adaptation benefit than an equivalent investment into capital intensive infrastructure. A significant component of agricultural community vulnerability remains structural in nature, and requires a policy solution to change behavior, and build awareness of best practice, both among policy makers and agricultural communities.

Adaptation Fund (AF) Additionality. With AF funding, the project covers the cost of communal water management arrangements through strengthening roles and capacities of local associations that will continue enforcing locally appropriate and tested adaptation measures in water access and management. Providing investment funds through Water User Associations will encourage capacity at community level delivery systems, and support their ability to engage with and leverage government social development funds. Over 30 000 people will benefit directly from the AF investment under this component. Although to date, government and donor support initiatives have achieved significant improvements in local land management practices, they have yet to achieve the scale-up to regional and national level as elaborated in this proposal. The project covers the cost of the replication of well proven adaptation measures through social protection and development programmes that will grant long term sustainability of the AF pilot investments.



All animal and vegetable beings consist of water: animals at 75%, fish at 75%, jelly-fish at 99%, potato at 76%, apples at 85%, tomatoes at 90%, cucumbers at 95% and melons at 96%. Even a human being is composed of water. There is 86% and 50% of water at any New-born and elderly bodies, respectively.



Partnership strategy and coordination between stakeholders

The following table sets out potential involvement by key institutions

Stakeholder name	Stakeholder mandate	Potential role in the project
Ministry of Nature Protection	Environment, Nature Protection, Climate Monitoring	Executing agency; main national implementing counterpart
Ministry of Agriculture	Land Use Planning, Distribution and Management of Arable Lands	Member of the Project Board
Ministry of Water Economy	Distribution and Management of Water Resources, Management and Development of Irrigation Infrastructure	Член совета проекта
Ministry of Economy	Economic Planning	Member of the Project Board
Research Institute of Water Management	Research on water quality and quantity issues	Project advisor
Institute of Desert, Flora and Fauna	Conservation and sustainable use of desert ecosystems and their resources	Project advisor
Institute for Strategic Planning and Development	Socio economic analysis; economic development trend and forecasting	Project advisor, recipient of modeling and prospective exercise training
Local Authorities	Local Planning and Administrative Decision-Making	Member of the Project Board
Local Communities	Use of Resources	Direct beneficiaries from the following regions: Karakum Region (Desert): Farmers and local authorities in several settlements of Rukhabat Etrap of Akhal Velayat Nohur Region (Mountainous): Farmers in the Garawul and Konegummez settlements , in Bakharly Etrap of Akhal Velayat Sakar-chaga region (Mary Velayat oasis): Dayhan Farm “Zachmet” (approximately 300 farmers)



Project goals and objectives

The main objective of *the project* is to strengthen water management practices at both local and national levels in response to climate change-induced water scarcity risks that are increasingly affecting farming systems in Turkmenistan. The project will assess and deliver concrete water adaptation measures to local vulnerable communities in the three typical agro-ecological regions, while also strengthening national level water legislation and pricing to ensure water availability for the non-state sector farmers. This combination of outcomes will ensure that concrete actions implemented through AF resources are sustainable beyond the lifetime of the project.

The programme of work has been structured to ensure that the various components are synergistic in terms of their objectives and delivery. The Government of Turkmenistan is already aware that water availability and affordability will be crucial to maintaining economic growth and social cohesion within the agricultural sector. This is particularly challenging when viewed against the projected climatic baseline set out in the Second National Communication for the UNFCCC (forthcoming). To meet this challenge, the government is focused on implementation of large scale water infrastructure investment designed to maximise storage and supply capacity. This is aimed at the support of large scale farming of strategic staple crops under the national procurement system (cotton, wheat, rice, sugar beet). Water efficiency infrastructure, such as sprinkler and drip irrigation systems, is also being integrated into these large scale farming operations. Even so, recent analysis undertaken within the UN IFF project, and findings within the Second National Communication indicate that such supply side measures are unlikely to be sufficient to meet the overall projected water deficit over the next 20 years. As a response, the government is issuing a new integrated national strategy for water and agriculture in 2011 which will address the efficiency issues in cotton and wheat farming.

The government therefore recognises that it is crucial to improve water availability from climate resilience and cost efficiency perspective, in particular for the significant proportion of the rural population that is engaged in small scale agriculture and livestock management. These populations exist outside of the national agro-procurement system, and in effect operate without state support. It is unlikely that the government will be able to meet the water needs of these communities through its infrastructure programmes, and as such, different approaches, including behavioural change, new low cost technologies, integrated planning and fiscal incentives will be required to ensure sustainability of these livelihoods as rain, river and groundwater fed systems become more challenged. Currently, the Government of Turkmenistan has limited experience of applying such demand side management approaches, and is seeking to identify effective ways to secure sustainable supplies for these marginal communities.

In this regard, there are a number of questions and challenges that the government faces that this project seeks to respond to in an integrated and comprehensive way. **These are:**

- What are the impacts of projected climate change on rural agricultural and livestock based communities within Turkmenistan who may not have access to large scale water infrastructure?
- What are the most effective technologies and management techniques for small scale rural demand management from a cost/benefit analysis, and how do these compare with supply side equivalents? Is it feasible to prioritize measures on a resource and cost efficiency basis?
- How do these water technologies and management techniques differ in terms of their suitability for different climatic and agricultural profiles within Turkmenistan and where are they best deployed?
- What are the most suitable delivery mechanisms for the provision, management and maintenance of sustainable water management systems?
- Can fiscal and billing mechanisms for water delivery be expanded to encourage more rational use of water by larger scale users without affecting poorer populations, thereby resulting in a more equitable allocation of water at a regional or river basin level?



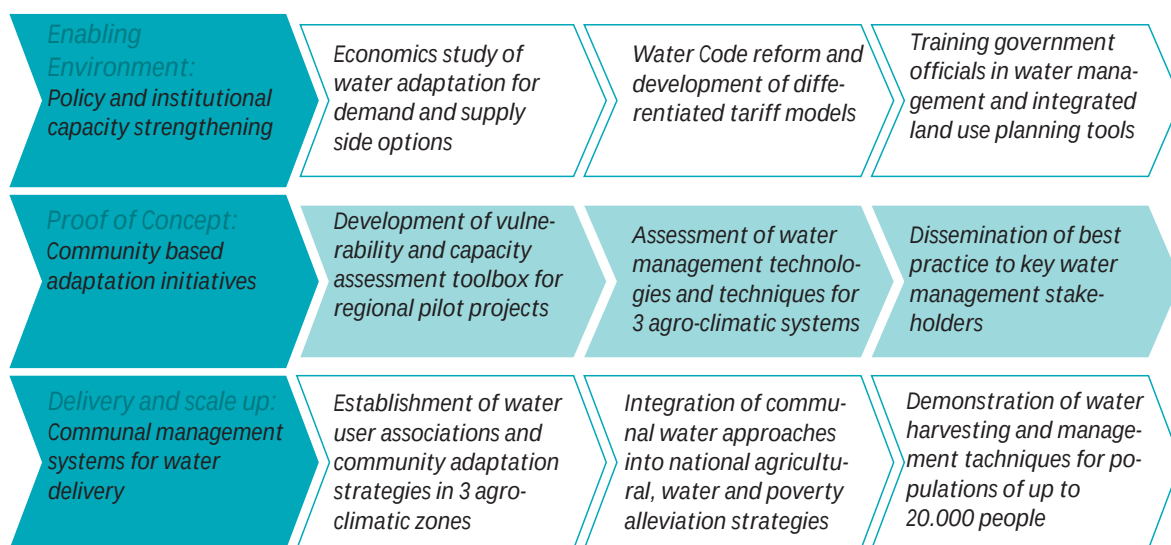
Annually precipitations averagely bring 12 tons of mineral salt per 1 km², and nearly 180 million tons of those ones fall to the ground in total. It is approximately a half of dissolved substances being annually carried by rivers to ocean.



- How can water and climate change considerations be integrated into agricultural sector and economic development planning, and what tools exist to facilitate this, particularly in relation to sustainability and resilience?
- Having identified potential solutions, what is the most effective way of scaling up community level best practice from local to national level, and how can knowledge be captured and replicated within national social development mechanisms?

To address these questions, the project is structured into 3 components as shown below:

Rationale for Turkmenistan Water Adaptation Programme



- **Component 1: Policy and Institutional Capacity Strengthening:** The first component is focused on improving the fiscal and management approaches by government towards water use in the state agriculture sector, in order to support the adoption of high efficiency irrigation techniques. This is important for local communities in that currently, water is diverted away from private sector agriculture and horticulture towards strategic state crops. In this regard, Component 1 seeks to build a sustainable basis for water adaptation, by weighing the cost effectiveness of various supply and demand side approaches, and by comparing these costs against those of the potential impacts of climate change, including from a social distribution perspective. This economic evidence base will be used to support water and agriculture modeling activities undertaken separately by the Ministries of Water Management and Agriculture. On the basis of economic outputs, it is expected that the project will support the reframing of water legislation to include climate change considerations, and help introduce regulations that support progressive water pricing and the communal management of water delivery services by the end of 2014. The linkages between water and potential reforms under the land use masterplan will also be established. It is not planned that there will be any market development activities for non-state crops as there is already well established demand for locally sourced vegetables, fruit and livestock products, and the issue is rather one of supply and productivity. The project will nonetheless work with any concurrent rural extension projects should they emerge to examine the potential to introduce added value processing and storage capacity to the communities involved in Component 2.
- **Component 2: Community based adaptation initiatives:** Recognizing that Turkmenistan has limited experience with community based adaptation approaches and a historic preference for adopting large scale supply side infrastructure solutions, it is considered vital that the project demonstrate the efficacy of the various community scale water adaptation approaches (water user



associations, drip irrigation, harvesting, water points, terracing, intercropping, saksaul planting, irrigation canal improvements etc.) A vulnerability assessment will be undertaken within 3 different agro-climatic regions-Nohur (Mountainous), Karakum (desert), and Sakar Chaga (Oasis) and hard water resilience measures introduced. These measures will primarily be targeted at community level approaches towards private sector agriculture, and the programme will seek to demonstrate the costs and benefits of these approaches at community level scale (up to 20 000 people). The lessons from these regional pilots will be used not only to inform the legislative reform process relating to land management and water use/pricing in component 1, but will also inform the development of larger scale communal management systems and their integration into the Government's social development and poverty alleviation strategy in Component 3.

- **Component 3:** Communal management systems for water delivery: The third component relates to the scale-up of community level water management and delivery mechanisms based around technologies and techniques proven in Component 2, and supported by the economic analysis and legislative changes set out in Component 1. The work of Water User Associations (WUAs) will be supported, and funds provided for WUA led community adaptation plans and concrete investments in water management systems and infrastructure in the 3 target regions. Investment funds will be provided to support at least 4 small scale investments. The project will support WUAs to improve water delivery services for the target communities as well as identify and develop suitable project ideas under their community based adaptation plans, and will select projects on the basis of impact (improvements in water quantity and efficiency). Match funding will be sought where feasible from government and other funds, thereby demonstrating leverage and building WUA capacity to diversify their sources of investment and income. The types of investments will include small scale water storage systems/dams, canal refurbishment investments, farm level water distribution, sustainable wells and ground water extraction, water level monitoring systems (levels and mineralization content), efficient sluice gates for field flooding, efficient communal pumping systems and efficient small scale pumps, water drainage and capture/reuse technologies, and integrated agronomy/water management systems (including fertiliser use, land consolidation, crop diversification). In addition to concrete investment planning, WUAs will be supported to develop mechanisms to support water efficiency systems. Aspects that might be considered include seasonal water rationing and more flexible payout mechanisms. Finally, the project will explore how achievements can be integrated into national level policy, and how best practice can be disseminated to other regions through the government, and through other channels.



Project output, components, and activities

The project is designed to contribute to the AF's portfolio level objective to «increase adaptive capacity to respond to the impacts of climate change, including variability at local and national level». The government of Turkmenistan requests a grant support from the Adaptation Fund (AF) to develop adaptive capacity of local communities in three agro-ecological zones of desert, oasis and mountainous parts (typical biophysical systems of the country). The government also requests support to develop policy incentives to increase water efficiency and help local communities adopt water related adaptation measures that will reduce water demand and improve supply services at the local level. Outcomes described below are aligned with the AF's outcome under objective 2 “Increased adaptive capacity within relevant development and natural resource sectors”.

Component 1:

Policy and Institutional Capacity Strengthening

Over the past 5 years, the government has invested approximately \$150-200 million in improving water delivery infrastructure (mainly in irrigation canals and drainage systems), and is planning to extend high efficiency irrigation technologies into the state agriculture sector. However, restructuring its water pricing policies is fundamental to achieving water efficiency and long term water productivity in the face of climate change. Decision-makers require the right type of information including rigorous risk assessments that incorporate socio-economic impact data. Currently, technical skill-sets and guidance are missing to undertake such assessments and to identify, design and implement sustainable climate resilient water policies. There are other knowledge gaps, including understanding of the requisite legal and regulatory foundations, economic costs, benefits and trade-offs and a host of other technical and feasibility issues. Through ongoing initiatives such as the Second National Communication and past initiatives such as the Investment and Financial Flows assessments, the government has acquired a certain degree of knowledge and understanding. However, more substantiated socio-economic arguments need to be put forward that will provide scientifically sound and evidence-based estimations of risks, costs of risks and cost-effective options to adapt. The following demonstrates the linkages between various subcomponents of regulatory reform and institutional capacity building:

- **Mapping the economics of water adaptation in the agriculture sector:** Currently, the choice of water management measures within government is not driven by a resource efficiency or cost-benefit perspective. There is a suspicion that current supply side measures represent a high cost response to potential water shortages in the agricultural system. The programme will finance a study, based on in-country fieldwork and a review of internationally available data, on the costs and benefits of various measures to manage water adaptation, both on the supply and demand side. This will make use of the emerging body of economic knowledge relating to adaptation, as well as the application of Integrated Assessment Modeling at a sector level. This will allow policy makers and communities to prioritize their technology and policy choices based on clear efficiency criteria, set against the cost of potential impacts in the sector.
- **Water Code reform:** Currently, Turkmenistan has a limited pricing structure for water, with charges for industrial and commercial agriculture, but with most social tariffs allowing free access to lower income users and state farmers. This results in poor and inefficient allocation mechanisms, and in a lack of incentives to undertake demand management by large portions of the agricultural community. On the basis of the outputs from the economics of water adaptation study, and the relevant sector model outputs, the project will seek to support changes to the Water Code to bring in a more progressive tariff structure. It will build capacity among relevant Ministries and authorities to design, for example, appropriate water tariffs that are socially sensitive and consider return value optimization. Change in water tariffs will yield improvements in water availability to poor farmers by raising the levels of water efficiency and freeing essential resources for better maintenance and improved water service delivery to the poor. Experience will be taken from international experience in



water reform, in particular pro-poor tariff structures being explored in South Asia, and by the UNDP's Human Rights based Water Management Programme in Central and Eastern Europe and CIS. Discussions have been undertaken already with the Ministry of Water Economy and the Parliamentary Committee that is responsible for the Turkmenistan Water Code, and there are indications that timing of the AF project will coincide with a potential update to the legislation. This component will also explore the linkages to the national land use master plan for the agriculture sector, including the farm distribution plan. Working with the Ministry of Agriculture and the Ministry of Economy, the project will explore the introduction of targets for increased water productivity and measures to achieve these.

Component 1 addresses the capacity building and reform requirement at national and regional government level to ensure that policy is evidence based (particularly integrating climate change projections), that water management techniques are implemented from a cost benefit perspective with Equal focus on supply and demand side approaches, and that the value of water is recognized as a commodity through progressive water pricing. These aspects build upon elements of existing reform and will feed through into land use policy. Without these, it is unlikely that demand side or community level approaches will be considered sensible by policy makers, and as such, these elements are considered vital. It is expected that activities under component 1 will create the enabling environment that will support effective community level water management and the shift towards Integrated Water Management that are developed in components 2 and 3.

Component 2:

Community-based adaptation initiatives.

The majority of AF resources will be allocated for implementation of concrete water and agricultural adaptation interventions at the community level. Despite increasing realization of water scarcity issues in Turkmenistan, and growing attention to and investments in water infrastructure, many communities that are not directly involved in cash crop production are unlikely to benefit from large scale investments. The majority of communities residing in three main agro-ecological zones of Turkmenistan are therefore under increasing pressure from water shortages. Their cropping, land and water management practices are often inappropriate in the context of pressures impinging upon the very ecosystem services they rely on. Consequently, land salinization, erosion and degradation of natural assets are common, thereby undermining the resilience of the communities in the context of emerging climate change risks. These pressures will be further amplified by prolonged droughts and overall aridification of the climate.

The project will support local communities with financial resources to design and implement adaptation measures that will help overcome water related limitations to their local productive systems. The project takes a bottom-up approach in assessing vulnerability and adaptive capacity in three geographic zones, representing three types of characteristic agro-ecological regions-mountain, desert and oasis-with agro-pastoralist, pastoralist and settled irrigated agriculture practiced, respectively. This type of geographic spread is determined by the potential for future replication of the measures that will prove successful in targeted compact localities and can be further scaled up in broader regions of similar characteristics.

Component 2 will begin by applying a **Vulnerability and Capacity Assessment (VCA)** to the identified communities which will serve as the basis for analysis of existing and potential hazards to the sustainable development of the agro-pastoralist system. The tool box will draw upon best practice from those existing VCA models that integrate climate resilience analysis as a core component, rather than those that focus primarily on disaster risk analysis. In particular, the following tools and frameworks (including the UNDP proprietary methodologies) will be assessed for local suitability in the context of the proposed programme.



Based on hydrologist estimations nearly 100 thousand km³ of water annually fall to the ground, each being of billion tons. 519 km³ of water fall to the ground per year. Those volumes of water make 42 “surface-to-air” rotations for the same period.



Nohur region (mountain region)

Agricultural background.

The region lies in the south-western part of Central Kopetdag Mountains that represent mountainous agro-ecological zones in the country, closer to the border with Iran. Inhabited by approximately 12 000 people, the region practices agro-pastoralism and is spread over 9 000 ha. Across the region, the natural Juniper forests were cut down to be used for heating purposes. Water scarcity is a problem due to low precipitation levels. Traditionally villagers collect run-off from mountain slopes and gorges in specially built reservoirs (howdams). In favorable years the howdams store sufficient water to irrigate the fields during one season. But during the last 5 – 6 years the how-dams have been left unfilled due to insufficient precipitation. In addition, the deforested slopes are failing to capture rainwater due to higher run-off levels. Due to water shortages, the population has gradually given up agriculture and horticulture, and has concentrated exclusively on livestock breeding. The pasture lands have as a result become overused, not letting the fodder for the animals re-grow on the stripped land, resulting in further erosion of the hill slopes.

Current Status.

Improved water harvesting and saving techniques are necessary to ease the increasing shortages, and to allow communities to revert to agriculture, make livestock management more sustainable and to stop the increasing degradation of the slopes. Some of the villages have already begun small scale activities. A number of small water infiltration dams have been constructed on the cleared slopes to increase the infiltration of the rainwater into the soil, as short-term solution. The dams constructed in ravines capture the water streams, which form during heavy rains, thus increasing the moisture in the soil and assisting in restoration of the vegetation. One village (Konegummez) is piloting drip irrigation systems for a 10 ha orchard with plans for a further 30 ha alongside reduction of livestock numbers. Additional activities have been undertaken to replant juniper trees to try and stabilize the soil and increase moisture retention with up to 100 ha of replanting. At a community level, a number of villages in the region have already begun work on Village Development Planning (VDP), and this process may be used going forward for the integration of climate change adaptation into community decision making. Close cooperation will be maintained with the ADB CACILM project, which looks at regional land management issues and which is considering climate change issues from a land management perspective in the region in the coming years.

Proposed Activities.

On the basis of the VCA, the project will work to develop and test water harvesting techniques, demand side efficiency measures and soil moisture management approaches that can support the livelihoods of 4,000 agro-pastoralists in the local communities. Techniques that will be evaluated and potentially implemented include water harvesting techniques, such as small scale slope terracing, contour stone and circular bunds, planting pits, living barriers. The rehabilitation of water infiltration dams on small streams will be explored. In terms of water retention and moisture capture perspective, the use of cover crops, mulching, minimum and zero tillage will be explored. To reduce evaporation and prevent loss of moisture, approaches may include use of windbreaks, dry and sparse seeding, fallow techniques, relay cropping and inter-cropping. From a water management perspective, further exploration will be undertaken of drip irrigation systems. The question of water pricing and provision of water services to off-grid communities will also be explored (building upon the water pricing work in Component 1).

Yerbent region (Karakum desert)

Agricultural Background.

Yerbent region is located in the Central Karakum Desert and occupies an area of almost 842 000 ha with 8,000 inhabitants. The relief of the region is a combination of dunes of various forms and takyr depressions. The greater part of the region is represented by desert pastures, and for centuries, the sand desert with a prevailing continental climate and high variability of precipitation has been used by



nomadic livestock breeders for extensive pasture. Intensive irrigation practices have contributed to land degradation. As people settled in villages, traditional practices of pasture management were forgotten. The lands started to be intensively exploited, especially around the villages and livestock water points. Saxaul bushes were dug out and used for heating and cooking purposes as other alternatives are either too expensive or not available, and have not recovered due to over grazing. Moving sand dunes, and more frequent heavy sandstorms not only threaten houses and other social infrastructure, they also degrade pasture land further and damage the limited number of existing wells. The degraded pasture lands do not produce sufficient fodder for the animals. The lack of watering points has led to the concentration of animals around existing wells. A structural change in the pastoral system (more goats and cattle instead of camels; more unguarded pastures around settlements instead of far pastures) has resulted in degradation of the vegetation, particularly around settlements and watering points. Community based well management and introduction of watering points are necessary.

Current Status.

A number of measures have been undertaken in relation to improved pasture management in the region that have proved successful. For example, Rukhabat Etrap has seen about 30 ha of reed reinforcement and 27 ha of saxaul planting to improve fixation of sand dunes. There have also been small scale repair and construction of water wells to expand the area of pasture land in use, thereby avoid localized degradation. Concrete water management requirements have been identified through the GTZ SLM project, and an opportunity exists to integrate water management into Village Development Plans (VDPs).

Proposed Activities.

On the basis of the VCA, the Karakum sub – component will develop and test community-based well and watering point management measures, as well as piloting traditional drought resistant grain varieties. It is envisaged that the sub – component will reach up to 8 000 farmers from the Yerbent region. The measures envisaged include fixation of sand dunes and water retention through saxaul planting and reed fixation. Focus will be upon scale up of water point availability through the upgrade of existing infrastructure and the development of new well infrastructure where appropriate. It is envisaged that 2 additional well points will have been constructed under the subcomponent by 2014. These will be undertaken on the basis of an environmental assessment and permissions from Turkmengeology and the Ministry of Nature Protection. Community level education and training will be undertaken to build awareness of sustainable livestock management practices.

Sakar-chaga region (Mary Velayat oasis)

Agricultural Background.

Sakar-chaga is located in the north-western part of Mary Velayat in the delta of Murgab River. The region occupies a total area of 53 000 ha. 34 settlements and 17 farmer associations are located in Sakar-chaga Etrap. Population of the region is 132 000, the largest part of which lives in oases where 80% of settlements are located. The region is considered to be a primary centre of arable farming in Turkmenistan for cotton, grain and vegetables. Soil salinization is the main problem of this site due to inadequate irrigation techniques and lack of drainage and this has resulted in very modest harvests and low yields. This has resulted in the abandonment of previously arable areas. The current irrigation management system leads to irrational use of both water and land resources. While water allowances are calculated centrally according to crop type, in practice, the provision of water is poorly controlled due to inadequate poor canal infrastructure, leading to over-irrigation in places, and a lack of water in others. In addition, there is evidence of an informal system of water payment practices that can favor some users over others.

Current Status.

A number of activities have been already undertaken in the region to stabilize agricultural output over recent years. Salinized soils are being regenerated and returned to productivity for small holders

through composting. In particular high quality humus is being produced to improve soil salinity. Sustainable land management techniques are being taught to young farmers. Restoration of degraded lands is being pursued in collaboration with local authorities, including bush clearance of 50 ha, ploughing, and desalination through preventive leveling. From a water management perspective, some initial activities are underway. Water user plans have begun to be elaborated for a small number of water users. From an infrastructure perspective, one new collector (3.3km) was constructed and an existing channel of 6.4l m cleared to improve collection of drainage water in the region. In addition, up to 60 measuring points have been installed to monitor the level and mineralization of ground water. However, these activities remain relatively small scale, and do not address expected increases in water vulnerability resulting from progressive climate change.

Proposed activities.

The Sakar-Chaga subcomponent focuses upon the strengthening the role of local associations in delivering irrigation services and introduction of a canal level management. The subcomponent envisages a pilot area of about 700 ha of irrigated lands provided with an upgraded infrastructure necessary to implement water-efficient irrigation plans. Local water users will be provided with the capacity to elaborate water efficient management schemes for the use of irrigation water. Lessons gained from decentralized water management planning by leaseholders will be elaborated documented and disseminated as part of the knowledge management strategy. In Sakar-chaga, the project may closely cooperate with a newly registered agricultural extension service which covers the topics of sustainable soil and efficient water management by i) individual consultation services; ii) Information days in the fields with farmers and iii) intensive training modules. This extension service will be registered by the end of 2010 and becomes operational by January or February 2011. The subcomponent will also explore water pricing and the development of water services from the perspective of local water users.

In summary, in the identified regions, Adaptation Fund (AF) resources will target socially vulnerable agricultural and pastoralist communities to implement the following concrete adaptation measures.

- On the demand side, a series of agronomic measures aimed at rebalancing increasing water demand will be implemented. For example, measures such as sand dune fixation/stabilization by planting local saksaul and other shrubs will be undertaken. Such measures are essential for moisture retention and revival of vegetation cover; terracing, intercropping and planting of drought resistant local varieties will minimize water demand for agricultural practices, including pastoralism, and support sustainable livelihoods.
- On the supply side, AF resources will be used to design and test improved water harvesting measures, internalization of climate change risks into well and irrigation canal management options and introduction of sustainable watering points for pastoralists.

The project will build upon existing community links developed under previous development projects, including the currently concluding GTZ Sustainable Land Management Project. Experience of working in these regions indicates that there is a high level of motivation to participate, given the marginal support received from the State. A number of community level organisations exist that will facilitate roll out of the envisaged analysis and measures. For example the Community Extension service being developed in Sakar-Chaga can be used for scale up and replication across the region. Commitments from the Parliament and respective national ministries indicate that the programme can expect regional support both to assist implementation, and to integrate best practice back into national strategy.

Research institutes, such as the Institute of Desert, Flora and Fauna and Research Institutes for Water Management under the Ministry of Nature Protection and the Ministry of Water Economy will be engaged in identifying and designing technical details of the above noted and other locally appropriate adaptation measures. The project will put in place sub-basin and/or irrigation canal level climate resilient management practices. One micro initiative has already been designed and implemented by

the ministry of Water Economy but further support is required. This will be done with AF resources by reviewing the earlier results of the Berzen pilot implemented by the Ministry of Water Economy.

Component 3:

Communal management systems for water delivery services introduced

Overview: Component 3 seeks to implement efficient communal water management systems in the selected regions, through the development of Water User Associations (WUAs) and the integration of communal water practices into wider social safety nets. Component 3 establishes the mandates and community investment plans for WUAs in the three identified regions. Investment funds will be provided to fund at least 4 water adaptation projects to be designed and implemented by WUAs, leveraging external government funds where appropriate. Lessons will be identified and scaled up to the national level.

Background: For the water related adaptation measures described under component 2 to be sustainable over longer periods of time, locally appropriate communal management structures will have to be put in place. This is to enhance the ability of communities to improve water delivery services, through improved management rights and greater accountability. Currently, the government is proposing large scale social programmes, designed to invest, largely in improving water and other physical and social infrastructure. The government also prioritizes social protection and development by focusing on water, namely upgrading drainage and irrigation infrastructure in the vast rural parts of the country. However, climate change risks and potential adaptation opportunities have not yet been accounted for in these programmes and they are often not accessible to small agricultural communities. Consequently, the long term sustainability of these investments is likely to be compromised. However, more farm-based, localized solutions are being overlooked, particularly for those farming systems that do not rely on irrigation. Important non-technological, communal management solutions for water productivity are necessary.

Activities:

- **WUA mandates and institutional capacity building:** Mandates and institutional functions of local water use associations in target regions will be strengthened to improve local water services that are more resilient to increasing water stress and benefit at least 30 000 farmers and pastoralists. Through the AF resources, these community associations will increase their functional role and engagement in shaping water adaptation solutions, including the delivery of water services. The project will strengthen existing community organizations/associations that will act as water supply and communal service providers. It is envisaged that the Water User Associations will support the pooling of resources for the operation and maintenance of their water system. The WUA will elect leaders, manage disputes, collect fees and implement maintenance, and will be built around water source infrastructure (either groundwater, water harvesting village dam or a canal). It is expected that the WUAs will operate within clearly defined resource boundaries, with an agreed set of rules, effective monitoring systems (water guards, gauges), and conflict resolution. Such structures increase accountability and more efficient water delivery and use.
- **Community based water adaptation plans:** Based on Vulnerability and Capacity Assessments (VCA) in each of the regions, community-based adaptation plans will be developed with particular focus on water delivery services. These will be designed and implemented in partnership with government social development programmes with direct engagement of at least 30 000 farmers and pastoralists. AF resources will support local associations and communal management organizations, such as water user associations, farmer associations and community organizations, to improve local response mechanisms and resilience to drought induced shocks and long term aridification that results in greater water shortages. Through a series of regular community mobilization meetings the communal associations will identify the most pressing needs in water



If all the glaciers thawed, the Earth water level would raise at 64 meters and nearly 1/8 of the land surface would be flooded.



access and services and in cooperation with the local government identify and plan the most acceptable solutions for subsequent funding from the social development programmes. Potential ideas may be cofunded under an investment programme through WUAs (below). Indeed, a key success factor of component 3 would be the inclusion of water related adaptation practices into the state social programmes that engage at a similar level and scale. This component aims to leverage a strategic opportunity to demonstrate the efficacy of integrating resilience into broader social vulnerability programmes.

- **WUA-led water sector investments at community level:** Finance will be provided for WUAs and associated community organizations to support investments in improved efficiency and quantity of agricultural water supply for local communities. On the basis of the above VCAs, the Water User Associations will prepare proposals for project financing. The project will support WUAs to formulate their investment proposals as part of their community adaptation plans under b. Potential investments will be selected on the basis of their ability to improve the delivery of water services, the ability to leverage match funding under government social protection or other funding mechanisms (see above), and the sustainability of the investment over time. At least 4 projects will be financed up to a total of \$ 400 000. This structure will encourage WUAs to build upon existing government social protection plans, and any emerging activities under the Agriculture and Water Strategies to 2030. It is expected that these structures will involve in excess of 30 000 farmers and water users across the 3 regions identified. The following list provides an overview of the potential types of investments that might be made by the Water User Associations:

- Small scale water storage systems/dams
- Canal refurbishment investments (clearance, lining, covering)
- Small scale distribution networks and farm level extension
- Sustainable wells and ground water extraction
- Water level monitoring systems (levels and mineralization content)
- Efficient sluice gates for field flooding
- Communal pumping systems and efficient small scale pumps
- Water infrastructure maintenance equipment
- Water drainage and capture/reuse technologies
- Integrated agronomy/water management systems (including fertiliser use), land consolidation, crop diversification

- **Lessons learned:** Lessons learned on community-based adaptation options in various agro-climatic conditions of Turkmenistan codified and disseminated (e.g. through ALM and other networks). Successfully tested adaptation measures under component 2 will be advocated for further replication and sustainable funding by empowered and capacitated community organizations that participate in this initiative. Community associations will be empowered to manage water harvesting infrastructure and climate risks to local distribution canals of central irrigation systems. This community based arrangement will allow for more equitable distribution of water. The project, through these communal management structures, will introduce such service innovations as seasonal water rationing and flexible payment options. By driving water service delivery to locally appropriate and self-sustained communal systems, water services will be sustained under the conditions of a changing climate. A well tailored hybrid of government and community managed services of water delivery will emerge as a more cost-effective solution compared to top down, subsidized and poorly maintained service. Regular lessons learned notes providing field-based experiences of local, community-based adaptation measures, improved preparedness and resilience of local livelihoods to drought and water shortages will be produced for dissemination.



MANAGEMENT ARRANGEMENTS

The Ministry of Nature Protection (MNP) is the government institution responsible for the implementation of the project and will act as the Executing Agency (EA). The Ministry of Water will be an implementing partner of component 1. At the request of the Government of Turkmenistan, UNDP is the Multilateral Implementing Entity (MIE).

As a Multilateral Implementing Entity, UNDP is responsible for providing a number of key general management and specialized technical support services. These services are provided through UNDP's global network of country, regional and headquarters offices and units and include assistance in: project formulation and appraisal; determination of execution modality and local capacity assessment; briefing and de-briefing of staff and consultants; general oversight and monitoring, including participation in reviews; receipt, allocation and reporting to the donor of financial resources; thematic and technical backstopping; provision of systems, IT infrastructure, branding, and knowledge transfer; research and development; participation in policy negotiations; policy advisory services; programme identification and development; identifying, accessing, combining and sequencing financing; troubleshooting; identification and consolidation of learning; and training and capacity building.

As outlined in UNDP's application to the Adaptation Fund Board for accreditation as a Multilateral Implementing Entity, UNDP employs a number of execution modalities determined on country demand, the specificities of an intervention, and a country context. Under the national execution modality proposed, UNDP selects a government entity as the Executing Entity based on relevant capacity assessments performed by UNDP. Please note that UNDP uses slightly different terminology to that used by the operational policies and guidelines of the Adaptation Fund. In UNDP terminology, the «executing entity» is referred to as the «Implementing Partner» in countries which have adopted harmonized operational modalities and the «Executing Entity» in countries which have not yet done so. The Executing Entity is the institutional entity entrusted with and fully accountable to UNDP for successfully managing and delivering project outputs. It is responsible to UNDP for activities including: the preparation and implementation of work plans and annual audit plans; preparation and operation of budgets and budget revisions; disbursement and administration of funds; recruitment of national and international consultants and personnel; financial and progress reporting; and monitoring and evaluation. As stated above, however, UNDP retains ultimate accountability for the effective implementation of the project.

The MNP will assume responsibility for the implementation, and the timely and verifiable attainment of project objectives and outcomes. It will provide support to the management unit, and inputs for, the implementation of all activities. The MNP will nominate a high level official who will serve as the National Project Director (NPD) for project implementation. The NPD will chair the Project Board / Project Steering Committee (PSC), and be responsible for providing government oversight and guidance to the implementation. The NPD will not be paid from project funds, but will represent a Government in kind contribution.

The project will be nationally executed by the Ministry of Nature Protection in cooperation with the Ministry of Water (mainly under component 1) that is the main institution responsible for integrating climate change policies in all sectoral strategies and local natural resource management systems.



Which kind of water would turn into ice hot or cold one? From the logical point of view it would surely be cold water. Hot water shall get cold at first and then be cold; while the cold one does not need getting cold. Experiments, however, prove that just hot water turns into ice faster.

National Execution enables a greater level of national ownership. UNDP will provide technical backstopping, quality assurance and compliance with fiduciary standards in its capacity of MIE.

A Project Steering Committee (PSC) will be convened by the MNP, and will serve as the project's coordination and decision-making body. The PSC meetings will be chaired by the NPD. It will meet according to necessity, but not less than once in 6 months, to review progress, approve work plans and approve major deliverables. The PSC is responsible for ensuring that the project remains on course to deliver products of the required quality to meet the outcomes defined. The PSC's role will include: (i) overseeing project implementation; (ii) approving all work plans and budgets, at the proposal of the Project Manager (PM), for submission to UNDP Regional Center in Bratislava; (iii) approving any major changes in plans or programmes; (iv) providing technical input and advice; (v) approving major deliverables; (vi) ensuring commitment of resources to support implementation; (vii) arbitrating any conflicts within the project and/or negotiating solutions between the project and any other stakeholders and (viii) overall evaluation.

The day-to-day administration will be carried out by a Project Management Team (PMT). As per Government requests, the staff will be recruited using standard UNDP recruitment procedures. The PMT will manage the implementation of all activities, including: preparation/updates of work and budget plans, record keeping, accounting and reporting; drafting of terms of reference, technical specifications and other documents as necessary; identification, proposal of consultants to be approved by the PSC, coordination and supervision of consultants and suppliers; organization of duty travel, seminars, public outreach activities and other events; and maintaining working contacts with partners at the central and local levels. The PMT will liaise and work closely with all partner institutions to link the project with complementary national programmes and initiatives. The PMT is accountable to the MNP, UNDP and the PSC for the quality, timeliness and effectiveness of the activities carried out, as well as for the use of funds. The PMT will produce Annual Work and Budget Plans (AWP&ABP) The PMT will further produce quarterly operational reports and Annual Progress Reports (APR) to the PSC, or any other reports at the request of the PSC. These reports will summarize the progress made versus the expected results, explain any significant variances, detail the necessary adjustments and be the main reporting mechanism for monitoring activities. The PMT will be technically supported by contracted national and international service providers, based on need as determined by the PMT and approved by the PSC. Recruitment of specialist services will be done by the PMT, in consultation with the UNDP and MNP and in accordance with UNDP's rules and regulations.



MONITORING FRAMEWORK AND EVALUATION

Project monitoring and evaluation will be conducted in accordance with established UNDP procedures. The logical framework provides performance and impact indicators for project implementation along with their corresponding means of verification. These and others to be developed prior to implementation will form the basis on which M&E of the project will be conducted. Table 6 provides a summary of the monitoring and evaluation plan plus a provisional budget. In order to have a realistic picture of impacts, outcomes and performance, as well as sustainability, it is important to know the perspective of local and national stakeholders. Therefore, stakeholders and selected communities will have a key role in the monitoring process. A crosssection of stakeholders will be associated to the monitoring of the project results. Stakeholder workshops featuring farmers, CBOs, local authorities, governmental and, and possibly nongovernmental organizations will regularly be carried out to monitor progress and disseminate results.

The views of farmers and their associations will be sought by questionnaire survey and group discussion, and those of Government administration by face – to – face dialogue. This will be completed by the project team's observations, to serve as a basis for analysis and reporting.

The objectives of Monitoring and Evaluation activities are:

- To analyze project progress, impacts and achievements
- To assess the relationship between activities planned in the project document and those implemented
- To re – orient the project, if needed
- To draw conclusions for future transfer of activities to other areas
- To allow exchange of experience with other projects within and out of the country

Before the start of implementation, an inception workshop shall be held with participation of the project team, relevant government counterparts, the UNDP-CO and others (civil societ representatives) etc.

This inception workshop will treat the following issues:

- The project's monitoring and evaluation plan.
- Fine – tuning of indicators, means of verification and assumptions. This will include reviewing the log frame
- Definition of M&E responsibilities of the project team
- First annual work plan of the project on the basis of the log frame matrix with precise and measurable performance indicators



Any thunder may be heard at 25 km away. And tension inside lightning is nearly 100 000 000 volt per meter.

Water reflects 5% of sunbeams, while snow does nearly 85%. Only 2% of sun light penetrates through the ocean ice.



The inception workshop will also provide an opportunity for all parties to understand their roles, functions, and responsibilities within the project's implementation process, including reporting and communication lines, and conflict resolution mechanisms.

Type of M&E activity	Responsible Parties	Time frame
<i>Inception workshop</i>	- Project Coordinator - UNDP-CO	<i>Within first two months of project start up</i>
<i>Inception Report</i>	- Project team - UNDP-CO	<i>Immediately following IW</i>
<i>Measurement of Means of Verification for Project Purpose Indicators</i>	- Project Coordinator	<i>Start, mid and end of project</i>
<i>Measurement of Means of Verification for Project Progress and Performance (measured on an annual basis)</i>	- Project Coordinator	<i>Annually prior yearly reports and to the definition of annual work plans</i>
<i>Monthly reports</i>	- Project team	<i>At the end of each month</i>
<i>Annual reports</i>	- Project team - CoRI - UNDP-CO	<i>At the end of each year</i>
<i>Meetings of the Project Coordination Committee</i>	- Project Coordinator - UNDP-CO	<i>After the inception workshop and thereafter at least once a year</i>
<i>Technical reports</i>	- Project team - External consultants	<i>To be determined by Project team and UNDP CO</i>
<i>Mid-term external evaluation</i>	- Project team - UNDP-CO - External consultants	<i>At the mid-point of project implementation</i>
<i>Final external evaluation</i>	- Project team - UNDP-CO - External consultants	<i>At the end of project implementation</i>
<i>Final Report</i>	- Project team - UNDP-CO	<i>At least one month before the end of the project</i>
<i>Publication of lessons learned</i>	- Project team	<i>Yearly</i>
<i>Audit</i>	- UNDP-CO - Project team	<i>Yearly</i>
<i>Visits to field sites</i>	- UNDP-CO - CoRI - Project team	<i>Yearly</i>

NOTE:

Above costs do not cover UNDP staff time. All UNDP staff costs associated with M&E are covered by the MIE Fee. The M&E budget will be taken pro-rata from the three project component budgets, reflecting the size of the TA.

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